

ACTION PLAN(by: LAKSHAYAA SUNDARARAJAN)

Airline Ticket Price Prediction using Machine Learning

Project Overview

Several factors like demand, travel season, booking period, route popularity, travel class, airline competition, and operational cost affect the price of airline tickets, which makes them change frequently. These prices are a complex mix of factors and predicting these prices manually is hard to do. The goal of this project is to build a machine learning model that can predict airline ticket prices, based on the available data.

The project adopts a clear machine learning process, from data collection to model evaluation and documentation. Various regression models are developed and tested for the most accurate prediction model. The final solution gives an example of using machine learning technologies to solve a practical business problem.

Objectives

The action plan is intended to guarantee the systematic completion of the project and guarantee proper documentation and accuracy of the model.

The objectives are:

- Compile an accurate airline ticket dataset.
- Understand the structure and characteristics of the dataset.
- Perform data cleaning and preprocessing.
- Perform exploratory data analysis (EDA).
- Carry out statistical analysis and hypothesis testing.

- Design relevant features to enhance predictions.
- Fit several multiple regression models.
- Evaluate models with evaluation measures.
- Choose the best performing mode.
- Store the learned model for future forecasts.
- Create project documents and presentation.
- Commit the completed project to GitHub as a repository to keep and share the project.

Project Scope

The project is centered around the prediction of airline ticket prices, based on data from past flights in the provided dataset. The commonly available flight data that is considered in the prediction model is airline, departure city, destination city, departure time, arrival time, number of stops, travel duration, travel class, and booking lead time.

This is an educational project and shows the full workflow of a supervised machine learning regression project. (It is not real time airline APIs nor continuously updating ticket prices.)

The project is divided into different phases to ensure the smooth execution of the project.

Phase 1 – Problem Identification

- Know and comprehend the problem.
- Define project objectives.
- Learn about the pricing of airline tickets.
- Identify suitable machine learning techniques.

Phase 2 – Data Collection

- Get the data from Kaggle.

- Load the data set into Jupyter Notebook.
- Inspect dataset dimensions.
- Examine feature descriptions.

Phase 3 – Data Preprocessing

- Remove duplicate records.
- Check missing values.
- Handle inconsistent data.
- Encode categorical variables.
- Verify data types.
- Create clean training data.
- The fourth phase is Exploratory Data Analysis.
- Analyse ticket price distribution.
- Analyze the price variation of the airlines.
- Compare travel classes.
- Examine journey duration.
- Analyse stop distribution.
- Generate correlation heatmap.
- Visualize important trends.

Phase 5 – Statistical Analysis

- Perform T-Test.
- Perform ANOVA.
- Analyse Pearson Correlation.
- Interpret statistical significance.

Phase 6 – Feature Engineering

- Encode categorical variables.

- Select important features.
- Remove unnecessary attributes.
- Create final feature matrix.

Phase 7 – Model Development

Train the following models:

- Linear Regression
- Random Forest Regression
- XGBoost Regression
- CatBoost Regression

Phase 8 – Model Evaluation

- For each model, use the following to evaluate:
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Make a comparison among all the models and choose the best algorithm.

Phase 9 – Documentation

Prepare:

- Research Report
- Action Plan
- Jupyter Notebook
- PowerPoint Presentation
- GitHub Repository

Project Timeline

Week Activities

- **Week 1:** Problem Statement, Data collection, Data features inspection, Data Cleaning and Data Pre-processing.
- **Week 2:** Perform exploratory data analysis, feature engineering, statistical analysis and hypothesis testing.
- **Week 3:** Train Linear Regression, Random Forest, XGBoost and CatBoost models. Evaluate the performance of the models.
- **Week 4:** Complete final model, document, prepare presentation, upload project to GitHub, submit.

Resources and Tools

- Hardware
 - Laptop
 - Minimum 8 GB RAM
 - Internet Connection
- Software
 - Python
 - Jupyter Notebook
 - Visual Studio Code
 - Python Libraries
 - Pandas
 - NumPy
 - Matplotlib
 - SciPy
 - Scikit-learn
 - XGBoost
 - CatBoost

- Joblib
- Dataset Source
- Airlines Flights Dataset on Kaggle.

Expected Deliverables

The project will generate the following outputs:

- Research Report
- Action Plan
- Jupyter Notebook (.ipynb)
- PowerPoint Presentation
- Trained Machine Learning Model
- GitHub Repository
- Project Documentation

Success Criteria

The project will be deemed a success if it is able to achieve the following aims:

- Successfully cleaned and preprocessed the dataset.
- Exploratory Data Analysis done.
- Correctly using statistical analysis.
- Multiple regression algorithms are implemented.
- Evaluation of model's performance based on the conventional measures.
- The optimum model is found and stored.
- All project documents are prepared.

Conclusion

This action plan is a structured roadmap for finishing Airline Ticket Price Prediction project. This workflow includes all aspects of the machine learning lifecycle, from data collection to documentation and submission of the project. This plan helps to ensure that all necessary steps are taken in a systematic and efficient manner, while keeping the ultimate prediction model accurate and reliable.